Chemistry GCE O Level 2019 - Paper 2

Duration: 2h30 | Cameroon GCE Board

CAMEROON GENERAL CERTIFICATE OF EDUCATION BOARD - ORDINARY LEVEL EXAMINATION - CHEMISTRY (0515), Paper 2, June 2019. Time allowed: two and a half hours.

This paper is arranged in three sections, A, B and C.

- Section A: answer 4 questions out of 5;
- Section B: answer 2 questions out of 3 and
- Section C: answer both questions.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage. Calculators are allowed.

You are reminded of the necessity for good English and orderly presentation in your answers.

Useful data:

Relative atomic masses	Hydrogen (H) = 1.0; Carbon (C) = 12.0; Oxygen (O) = 16.0; Copper (Cu) = 64.0
1 Faraday	96000 coulombs
Molar volume of a gas at r.t.p.	24000 cm^3
Specific heat capacity of water	4.2 J g^{-1} $^\circ C^{-1}$
Avogadro number	6.02×10^{23}
$0^\circ C$	273 K

Exercise 1 (10 marks)

This question concerns the first four elements in group I of the periodic table.

Element	Lithium (Li)	Sodium (Na)	Potassium (K)	Rubidium (Rb)
Atomic number	3	11	19	37

a. (i) Why are they classified as Group I elements? *[1 mark]*

(ii) Which of the elements is the least reactive? *[1 mark]*

(iii) Write the electronic configuration of potassium (K). *[1 mark]*

b. (i) These elements are usually stored under paraffin. Explain.

(ii) Write an equation for the reaction between sodium and water. *[3 marks]*

c. Lithium though in Group I of the periodic table resembles magnesium in Group II in its chemical properties.

(i) What name is given to this resemblance?

(ii) State the products formed when the carbonates of Li and Na are strongly heated:

$\mathrm{Li_2CO_3}$

$\mathrm{Na_2CO_3}$ *[3 marks]*

d. State the type of compound formed when Group I elements combine with non-metals. *[3 marks]*

Exercise 2 (10 marks)

Aluminium is prepared industrially from purified bauxite by electrolytic reduction.

a. (i) Write the formula of the oxide.

(ii) Write the cathode half reaction for the process. *[2 marks]*

b. (i) Can chemical reduction be used to extract Al from its ore?

(ii) Explain. *[2 marks]*

c. (i) Why are Al extraction plants usually located near power stations? *[1 mark]*

d. Give one large scale use of Al and relate this use to its property.

Use:

Property: *[2 marks]*

e. (i) State one metal that can be extracted by chemical reduction.

(ii) Name an ore from which this metal is extracted.

(iii) Write an equation for the extraction process. *[3 marks]*

Exercise 3 (10 marks)

A homologous series of organic compounds has the general formula $\mathrm{C_nH_{2n+1}OH}$.

a. (i) What is a homologous series?

(ii) Give the general name of this homologous series.

(iii) Name the second member of this series. *[3 marks]*

b. Members of the above series can react with metals.

(i) Write a balanced equation to show how the second member of the series reacts with sodium.

(ii) Name the non-gaseous product formed in (b)(i) above. *[2 marks]*

c. The second member of the series reacts with ethanoic acid.

(i) What name is given to this type of reaction?

(ii) Write an equation for the reaction.

d. State one simple test for members of this homologous series. *[1 mark]*

e. State two large scale uses of the second member of this homologous series. *[2 marks]*

Exercise 4 (10 marks)

This question concerns the laboratory preparation of ammonia ($\mathrm{NH_3}$) by reacting a mixture of ammonium chloride and calcium hydroxide.

a. (i) Why is it advisable to grind the mixture of ammonium chloride and calcium hydroxide?

(ii) State the reaction condition. *[2 marks]*

b. Write an equation for the reaction. *[1 mark]*

c. State the drying agent used. *[1 mark]*

d. Ammonia gas is used as a reducing agent for copper(II) oxide.

(i) Write an equation to show the reduction of $\mathrm{NH_3}$ [as printed; the equation required is the reaction in which ammonia reduces copper(II) oxide].

(ii) State one observation when the copper(II) oxide is reduced. *[3 marks]*

e. Give one large scale use of

(i) Nitrogen.

(ii) Ammonia. *[2 marks]*

Exercise 5 (10 marks)

Sulphur dioxide reacts with air according to the equation below:

$$2\mathrm{SO_2(g)} + \mathrm{O_2(g)} \rightleftharpoons 2\mathrm{SO_3(g)}\ \Delta H = -196\ \mathrm{kJ/mol}$$

a. (i) Explain the meaning of $\Delta H = -196$ kJ/mol.

(ii) State two (2) reaction conditions for the above reaction. *[3 marks]*

b. (i) State Le Chatelier's principle.

(ii) What will happen to the equilibrium position when

- pressure is increased?

- temperature is increased? *[3 marks]*

c. What volume of oxygen is required to produce 300 cm^3 of SO_3 at room temperature and pressure? *[2 marks]*

d. State one environmental hazard of SO_2. *[1 mark]*

e. Give one large scale use of SO_3. *[1 mark]*

Exercise 6 (20 marks)

Section B: answer any two questions. All questions carry equal marks. Where appropriate, equations and diagrams should be used to illustrate your answer.

Bonding in substances can either be ionic, simple covalent or metallic.

a. Define chemical bond. *[1 mark]*

b. Using appropriate diagrams, describe how each bond type occurs in a named substance. *[15 marks]*

c. Give one property of each substance, stating how this property is related to the bond type. *[4 marks]*

Exercise 7 (20 marks)

Section B: answer any two questions. All questions carry equal marks. Where appropriate, equations and diagrams should be used to illustrate your answer.

a. Define the following terms.

(i) Solubility

(ii) Solubility curve *[3 marks]*

b. The table below gives the solubility of potassium nitrate (KNO_3) in grams per 100 g of water at different temperatures.

Temperature in $^\circ$C	**0**	**10**	**20**	**30**	**40**	**50**	**60**
Solubility of KNO_3 in g per 100 g of H_2O	16	21	31	45	62	81	106

Plot the solubility of KNO_3 against temperature (X-axis). *[10 marks]*

c. From the graph determine

(i) the solubility of the salt at 45 $^\circ$C;

(ii) the temperature at which 40 g of KNO_3 will saturate 100 g of water. *[2 marks]*

d. (i) What mass of KNO_3 would be obtained when a saturated solution is cooled from 50 $^\circ$C to 15 $^\circ$C?

(ii) Calculate the amount of substance of KNO_3 in mol found in 100 g of water in (d)(i) above. *[4 marks]*

e. State one use of solubility curves. *[1 mark]*

Exercise 8 (20 marks)

Section B: answer any two questions. All questions carry equal marks. Where appropriate, equations and diagrams should be used to illustrate your answer.

Write short notes on each of the following.

a. Polymorphism *[5 marks]*

b. Saponification *[5 marks]*

c. Isotopy *[5 marks]*

d. Hydrolysis *[5 marks]*

Exercise 9 (20 marks)

Section C: answer all questions in this section.

In order to determine the concentration of a dilute sulphuric acid solution, a student is provided with the following solutions and equipment: 0.25 M NaOH, dilute HCl, pipette, burette, conical flask, phenolphthalein indicator, clamp and stand.

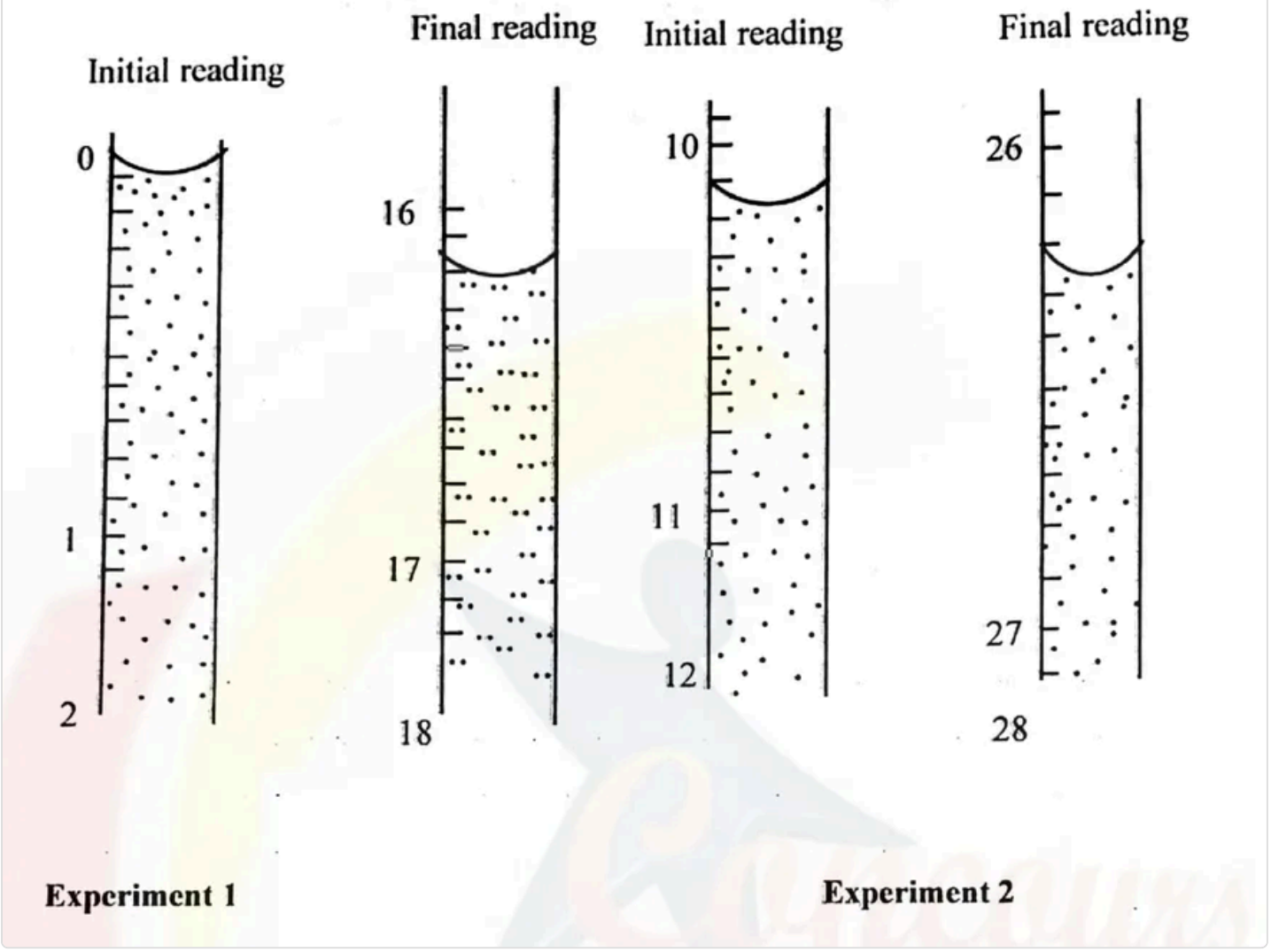
a. Draw the experimental set up you would use. *[3 marks]*

b. 25 cm^3 of 0.25 M NaOH was transferred into a conical flask and 3 drops of phenolphthalein indicator was added.

(i) Sketch and name the apparatus used to transfer the NaOH solution.

(ii) Identify the liquid that is used to rinse the conical flask. *[3 marks]*

c. The dilute acid was run from the burette and two accurate results were recorded as shown on the following diagrams:



(i) Read and record the results of experiment 1 and 2 on the following table. *[3 marks]*

	Experiment 1	**Experiment 2**
final reading		
initial reading		
Titre		

(ii) Determine the mean titre of the dilute sulphuric acid used. *[1 mark]*

(iii) What will be the colour change to indicate the end of the experiment? *[1 mark]*

(iv) Calculate the concentration of the dilute sulphuric acid solution if 25 cm^3 of 0.25 M NaOH was required. *[2 marks]*

(v) In carrying out this experiment the acid accidentally spilled on the student's hands. Advise the student on what should be done. *[1 mark]*

d. You are given the following gases: H_2, SO_2, NH_3.

Sketch the apparatus you would use to collect a pure sample of each gas in the laboratory. *[6 marks]*

Exercise 10 (20 marks)

Section C: answer all questions in this section.

In order to determine the chemical composition of three unknown substances X, Y and Z, a student carried out the following tests.

a. Study the following table and complete the information needed.

No	Test procedure	Observation	Inference	Marks
(i)	To 2 cm^3 of solution X is added 5 drops of NaOH(aq), warmed	A colourless pungent gas is evolved. The gas turns damp red litmus paper blue	.....	2 marks
(ii)	To 2 cm^3 of solution X is added 2 drops of BaCl(aq) followed by dil HCl(aq)	.....	Presence of SO_4^{2-} ions	2 marks
(iii)	To 2 cm^3 of solution Y is added 2 drops of NaOH(aq)	.....	Presence of Fe^{2+} ions	1 mark
(iv)	To 2 cm^3 of solution Y is added a few drops of $AgNO_3$ followed by dil HNO_3	A white ppt is formed	.....	1 mark
(v)	.....	Z gives a brick red flame colour	.....	3 marks
(vi)	A solid sample of Z strongly heated in a test tube	.....	NO_2 evolved; presence of NO_3^-	1 mark

b. Give the chemical identity of

X

Y

Z *[3 marks]*

c. Complete the following table to show how you will separate these mixtures. *[4 marks]*

Mixture	Technique of separation
Kerosine and water	.....
.....	Simple distillation
.....	Paper chromatography
Groundnut and its peelings	.....

d. (i) In order to dilute an acid, a student added 410 cm^3 of distilled water to 10 cm^3 of concentrated sulphuric acid. State and justify what is wrong with this procedure.

(ii) Why is it advisable to wear eye goggles when working in the laboratory? *[3 marks]*

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

View the correction

Scan me

edukora.africa - Free-access paper, feel free to share. The correction is for subscribers only.